



# **RIT 1201 AI TEAM 9**

## **Exploratory Data Analysis and Visuaization Report**

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### Team Members:

- |                         |                         |
|-------------------------|-------------------------|
| 1. Muhammad Subhan Khan | 5. Swati Mani           |
| 2. Muhammad Tafoor      | 6. Omotosho Johnson     |
| 3. Sania Farhen         | 7. Olawale Owoyomi      |
| 4. Helen Ndarake        | 8. Sasikanth Danthanala |

Date of Submission: 26/1/2026

## **1. Introduction:**

### **1.1 Dataset Overview**

This report is based on a cleaned and preprocessed dataset prepared during the initial phase week1 of the internship. The dataset contains opportunity-wise user activity records, primarily focusing on signup behavior and completion outcomes over time.

The dataset is structured in a tabular format, where each row represents an individual user or opportunity record. Key columns include signup date, completion date, completion status, completion duration (in days), and additional derived time-based features such as signup day, weekday, and month. These engineered features enable detailed time-series analysis and behavioral trend identification.

The dataset is suitable for exploratory analysis as it captures both user intent (signups) and user follow-through (completions), which are critical indicators of engagement and conversion effectiveness.

### **1.2 Analysis Goals**

The primary objective of this analysis is to understand patterns, trends, and anomalies in signup and completion behavior. Specifically, this analysis aims to:

- Identify growth trends and fluctuations in signups over time.
- Examine seasonal patterns and recurring behaviors.
- Evaluate the stability and efficiency of completion outcomes.
- Detect outliers and abnormal behaviors that may indicate friction or process gaps.
- Derive actionable insights that can support business decision-making.

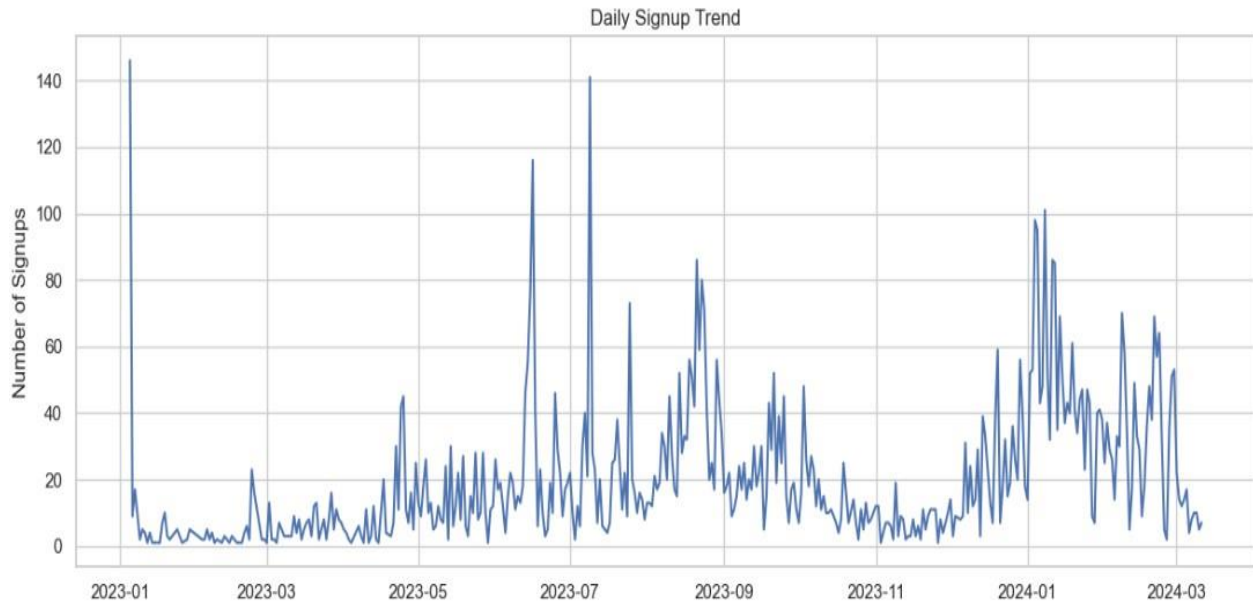
Understanding signup and completion trends is essential for business success because signups indicate interest, while completions reflect value realization. A gap between these two metrics often highlights engagement issues, onboarding inefficiencies, or mismatched expectations.

## **2. Signup Trends Analysis:**

### **2.1 Signup Growth Over Time**

To evaluate overall growth, daily signup counts were aggregated and visualized using line charts. This approach allows clear identification of long-term trends, short-term fluctuations, and directional momentum.

The visualization reveals that signup activity does not follow a perfectly linear growth pattern. Instead, it exhibits phases of acceleration and slowdown. Periods of increased signup volume may correspond to campaigns, platform updates, or external events, while declines may indicate reduced visibility or user fatigue.



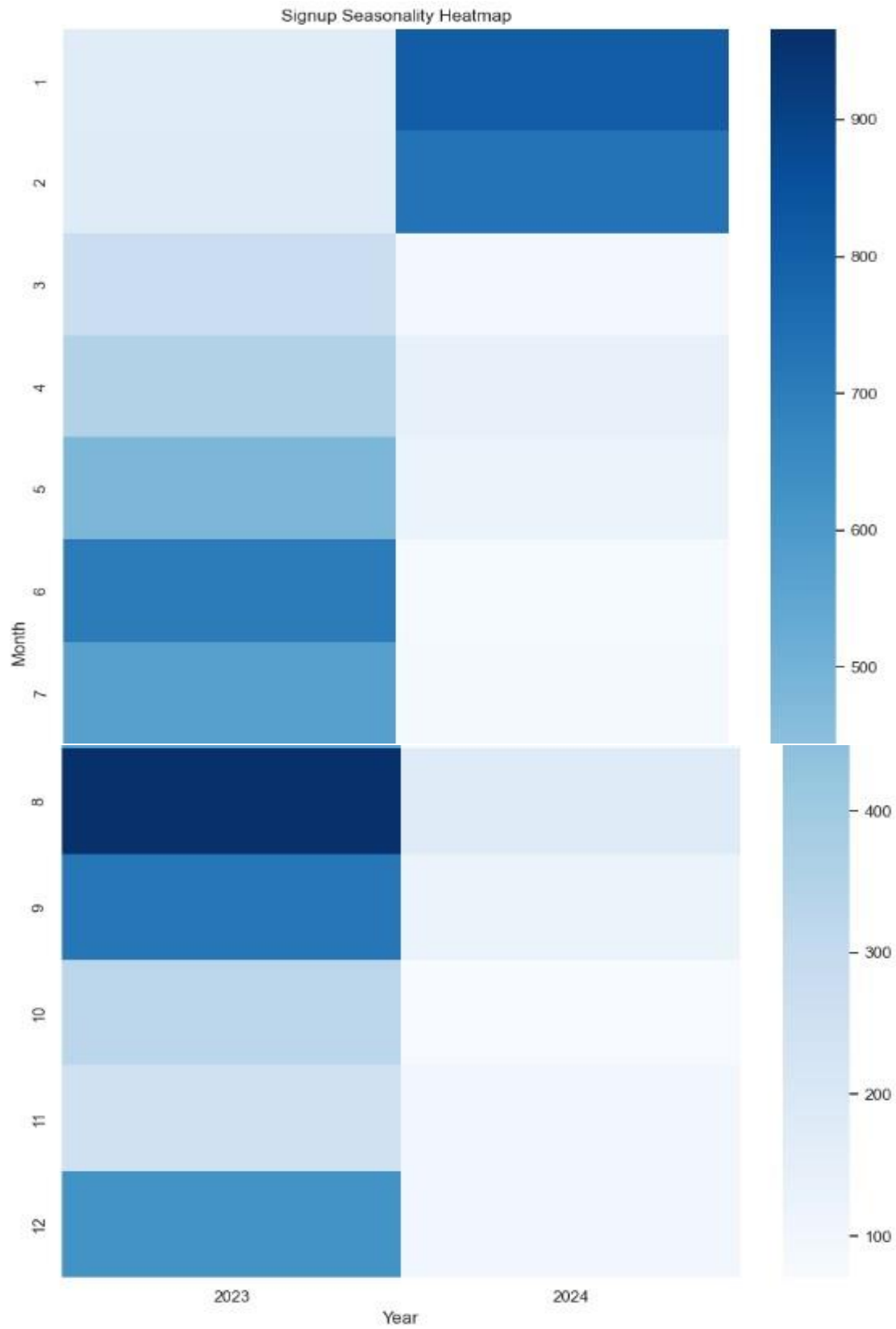
*Line Graph: Daily Signup Growth Over Time*

## 2.2 Seasonality Patterns

Seasonality analysis was conducted by examining signup activity across weekdays and months using heatmaps and time-series plots. This approach helps identify recurring temporal patterns that are not immediately visible in raw trend lines.

The heatmap analysis highlights consistent peaks during specific weekdays and months, suggesting that user behavior is influenced by temporal factors such as work schedules, academic calendars, or marketing cycles. Certain months consistently show higher engagement, while others experience noticeable drops.

These seasonal insights are valuable for planning marketing efforts, resource allocation, and campaign timing.



*Heatmap: Signup Seasonality (Month vs Year)*

## 2.3 Signup Spikes and Drops

Further analysis focused on identifying days with unusually high or low signup counts. These spikes and drops were examined in context rather than treated as random noise.

Significant spikes may indicate successful promotional efforts or external exposure, whereas sharp drops could be caused by technical issues, reduced outreach, or competing events. Identifying these days allows stakeholders to investigate underlying causes and replicate successful conditions while mitigating negative factors.

```
++++ Signup Spikes and Drops +++++
```

```
Top Signup Days:
```

```
  Learner  SignUp  DateTime
```

```
2023-01-05      146
```

```
2023-07-09      141
```

```
2023-06-16      116
```

```
2024-01-08      101
```

```
2024-01-04       98
```

```
dtype: int64
```

*Table: Top 5 Highest Signup Days*

```
Lowest Signup Days:
```

```
  Learner  SignUp  DateTime
```

```
2023-04-15       1
```

```
2023-02-19       1
```

```
2023-11-03       1
```

```
2023-02-14       1
```

```
2023-02-11       1
```

*Table: Lowest 5 Signup Days*

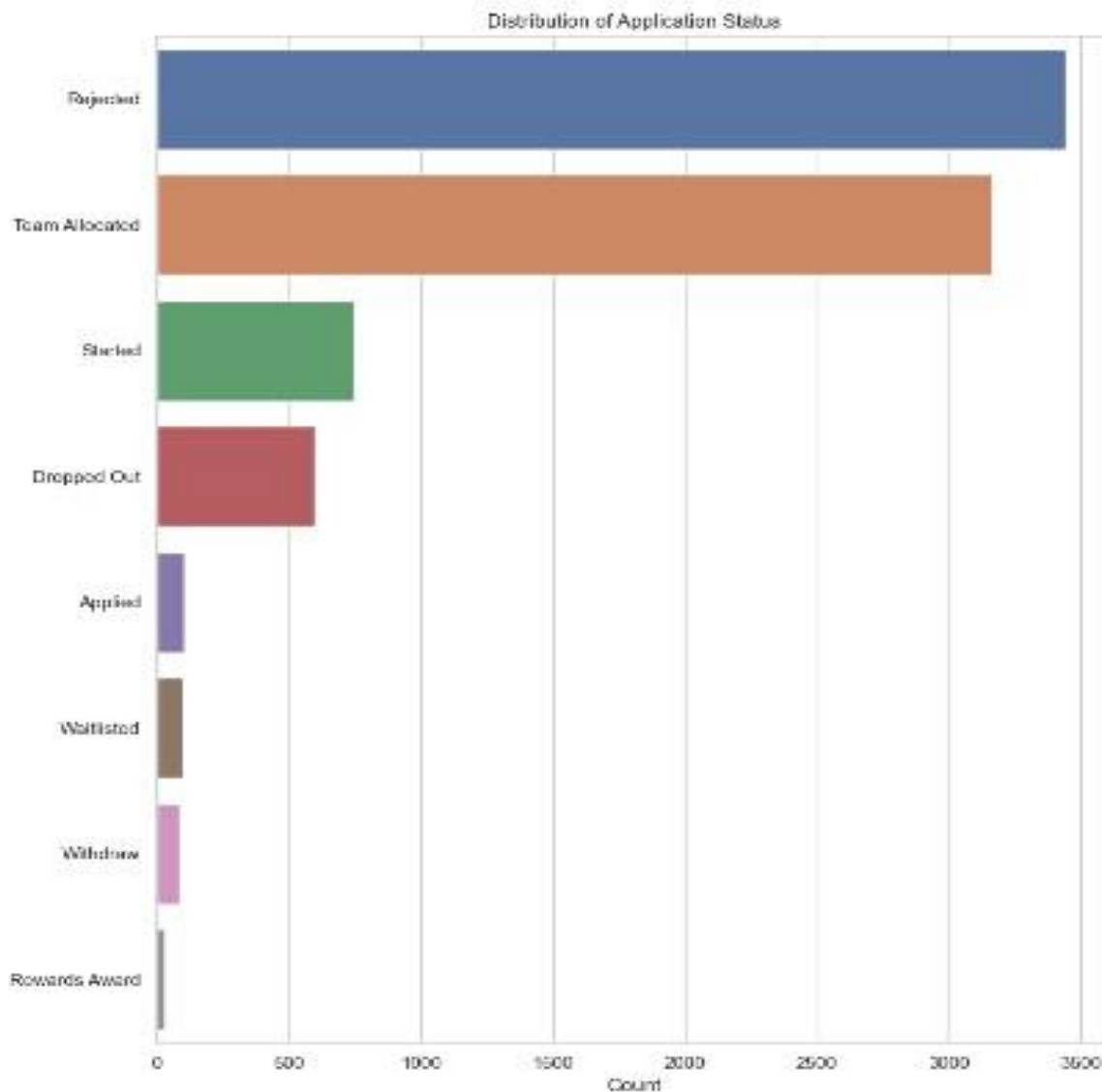
### 3. Completion Trends Analysis:

#### 3.1 Completion Rate Stability

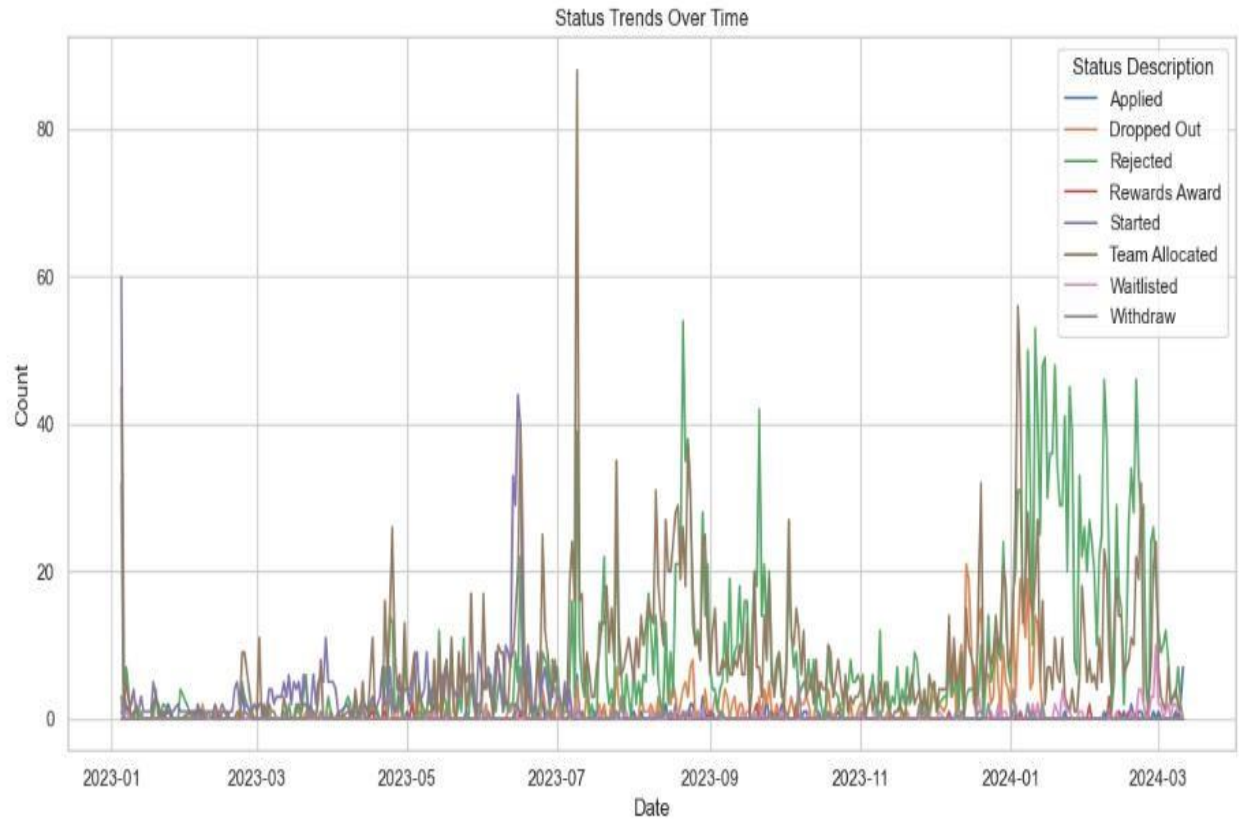
Completion rates were calculated as the proportion of completed records relative to total signups on a given day. Line charts were used to visualize how completion rates evolve over time.

The analysis shows that while completion rates remain relatively stable overall, certain periods experience noticeable dips. These dips may indicate onboarding challenges, unclear requirements, or reduced user motivation during specific timeframes.

Monitoring completion rate stability is crucial because consistent signups without corresponding completions can negatively impact overall platform effectiveness.



*Count Plot: Distribution of Application Status*



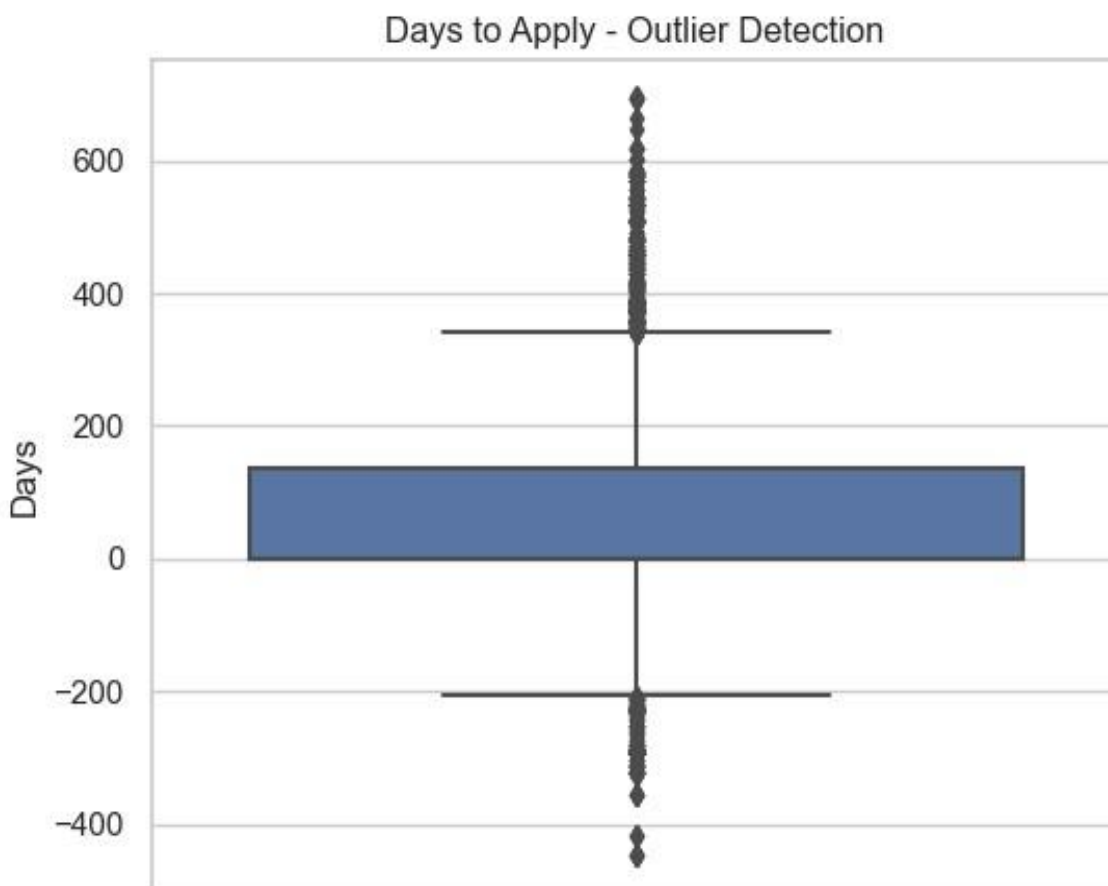
***Line Chart: Completion Rate Trend Over Time***

### 3.2 Completion Time Variations

Completion time, measured in days between signup and completion, was analyzed using box plots to understand distribution, spread, and variability.

The box plot reveals that most users complete within a typical timeframe, indicating a generally efficient process. However, the presence of outliers suggests that a subset of users experiences significantly longer completion times.

These variations may reflect differences in user commitment, task complexity, or support availability.



*Box Plot: Completion Time Distribution*



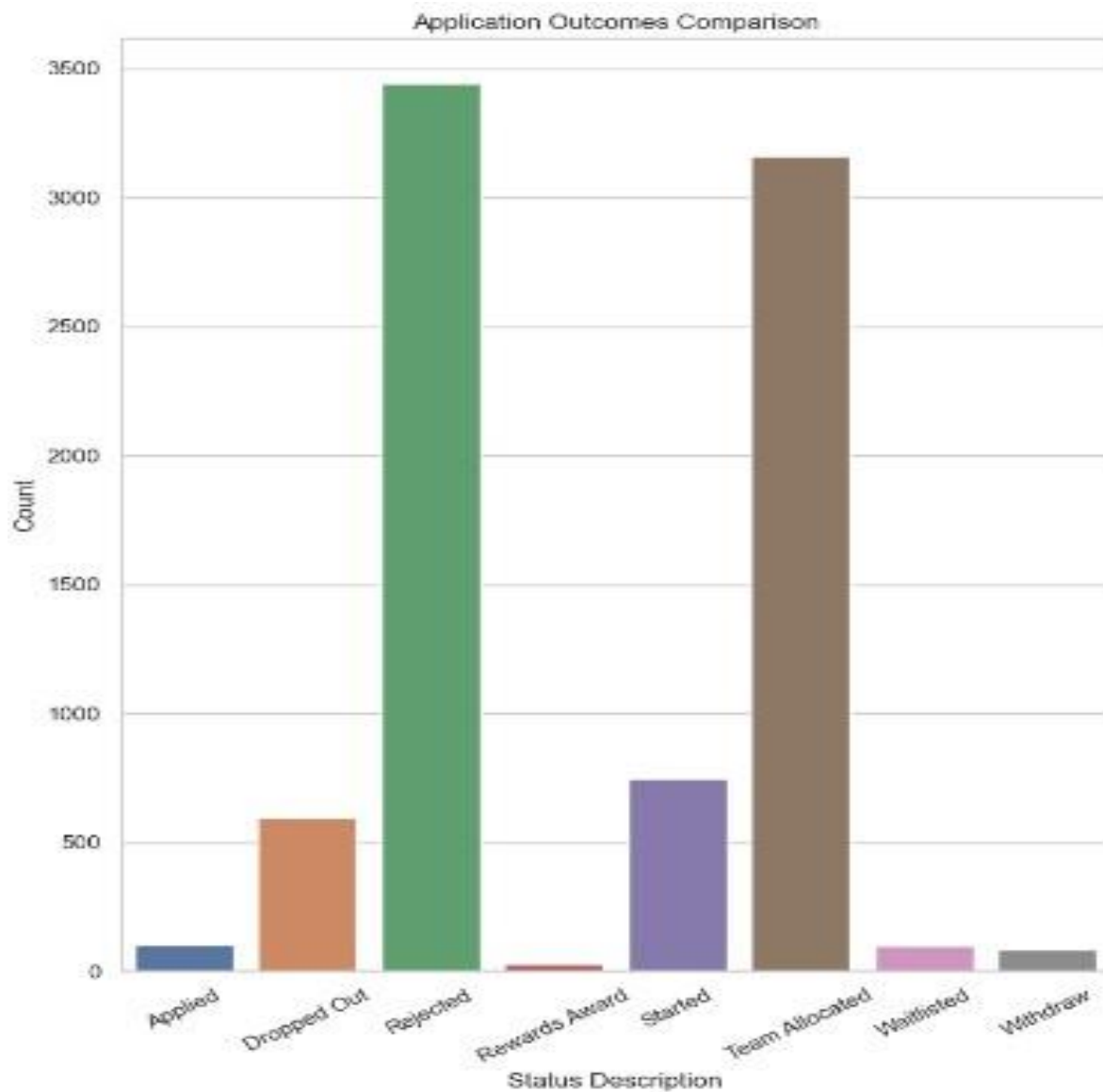
## 4. Patterns and Correlations:

### 4.1 Relationship Between Signups and Completions

To explore whether higher signup volumes translate into higher completions, a scatter plot was used to visualize the relationship between daily signups and daily completions.

The analysis shows a positive correlation, indicating that increased signups generally lead to increased completions. However, the relationship is not perfectly proportional. On certain days, high signup volumes result in relatively low completion counts, highlighting potential drop-offs in the user journey.

This finding suggests that scaling signups alone is not sufficient; equal focus must be placed on improving conversion and engagement mechanisms.



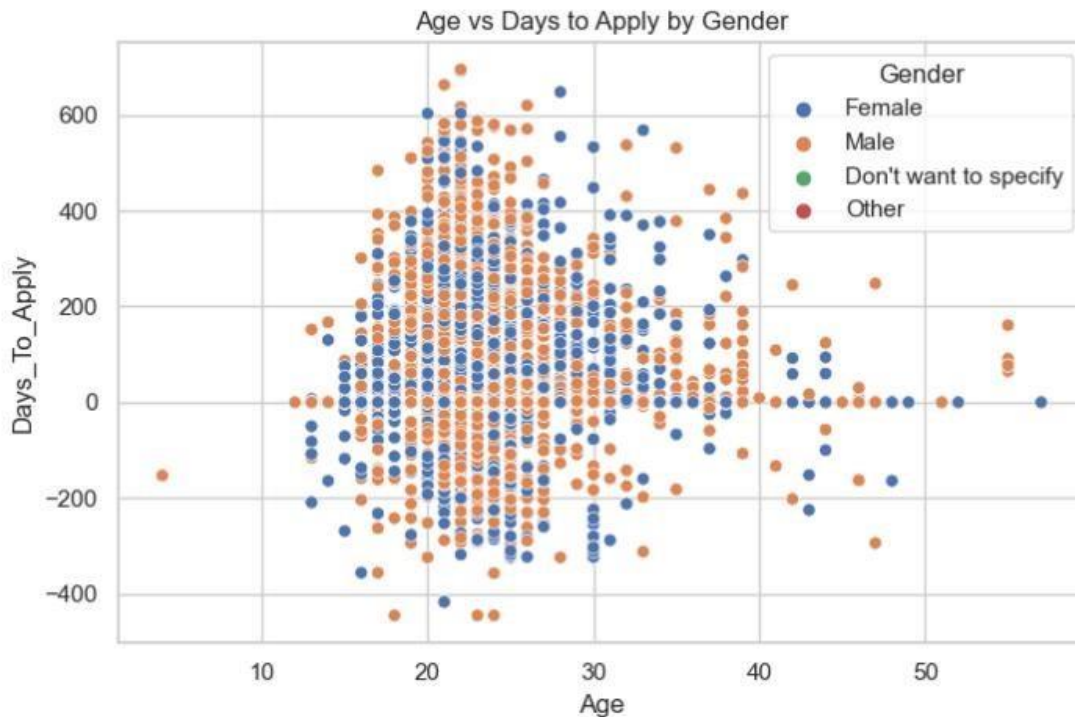
*Bar Plot: Signups vs Completions*

## 4.2 Demographic and Segment-Based Analysis

Where demographic or categorical data is available, performance was analysed across different user groups. This segmentation helps identify which groups exhibit higher completion rates and faster completion times.

Certain segments consistently outperform others, suggesting that tailored communication, support, or onboarding strategies may improve outcomes for underperforming groups.

Segment-based insights enable targeted interventions rather than one-size-fits-all solutions.



*Scatter Plot: Age vs Days to Apply by Gender*

## **5. Outliers and Anomalies:**

### **5.1 Completion Time Outliers**

Outliers in completion time were identified using interquartile range (IQR) analysis. These records represent users who took significantly longer than the majority to complete.

Such outliers may indicate users facing difficulties, lack of clarity, or reduced motivation. While they represent a small proportion of the dataset, addressing their challenges could improve overall completion efficiency.

### **5.2 Low Completion Days**

Days with unusually low completion counts were identified and analysed in relation to signup volume. In some cases, low completions coincided with low signups, while in others, they occurred despite moderate signup activity.

This discrepancy suggests potential operational issues, such as system downtime, delayed communication, or user disengagement.

## **6. Recommendations:**

### **6.1 Target Peak Signup Days**

Based on observed seasonality and peak activity periods, marketing and outreach efforts should be aligned with high-performing days and months to maximize return on investment.

### **6.2 Investigate Signup and Completion Drops**

Sharp declines in signup or completion metrics should be proactively investigated through crossfunctional analysis involving marketing, product, and support teams.

### **6.3 Support Long-Tail Users**

Users with extended completion times may benefit from additional reminders, onboarding assistance, or resource availability. Targeted support can reduce drop-offs and improve overall efficiency.

### **6.4 Segment-Based Strategies**

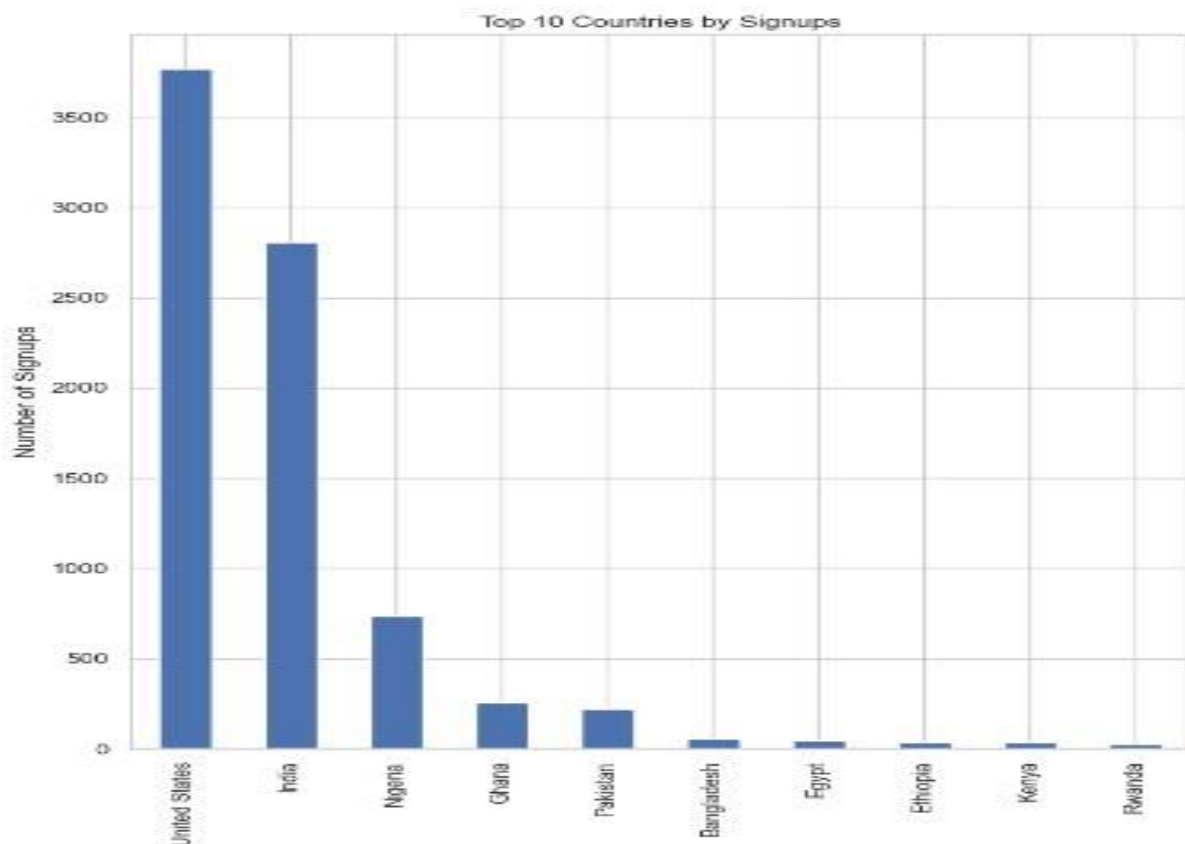
Different user groups exhibit different engagement patterns. Tailored strategies such as customized messaging, incentives, or guidance can significantly improve outcomes across segments.

```
# Outliers & Anomalies:
print("++++ Long-Tail Users (Very High Days to Apply) +++++")
Q1 = df['Days_To_Apply'].quantile(0.25)
Q3 = df['Days_To_Apply'].quantile(0.75)
IQR = Q3 - Q1

outliers = df[df['Days_To_Apply'] > Q3 + 1.5 * IQR ]

print("Number of long completion cases:", outliers.shape[0])
```

```
++++ Long-Tail Users (Very High Days to Apply +++++
Number of long completion cases: 460
```



*Bar Plot: Country-wise Signup Analysis*

## 7. Conclusion

### 7.1 Key Insights

- Signup activity shows clear growth patterns with identifiable seasonal trends.
- Completion rates are generally stable but exhibit periodic declines.
- Completion time varies, with a small number of significant outliers.
- Higher signups do not always guarantee proportional completions.

### 7.2 Next Steps

Future analysis can include cohort-based retention studies, predictive modeling for completion likelihood, and deeper segmentation to further optimize user engagement and conversion strategies.

#### Final Notes for Submission

- All visualizations were generated using Python (Pandas, Matplotlib, Seaborn).
- Code execution screenshots should be attached at the end of relevant sections.
- The cleaned dataset serves as the analytical foundation for all findings.